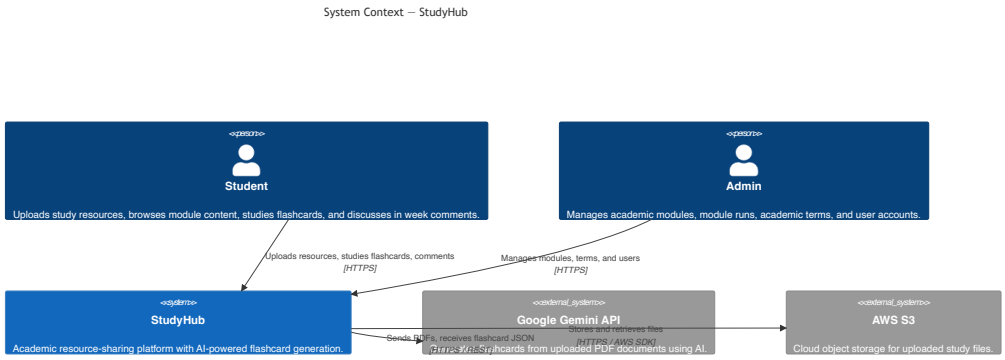
StudyHub — C4 Architecture Diagrams

System architecture across Context, Container, and Component levels

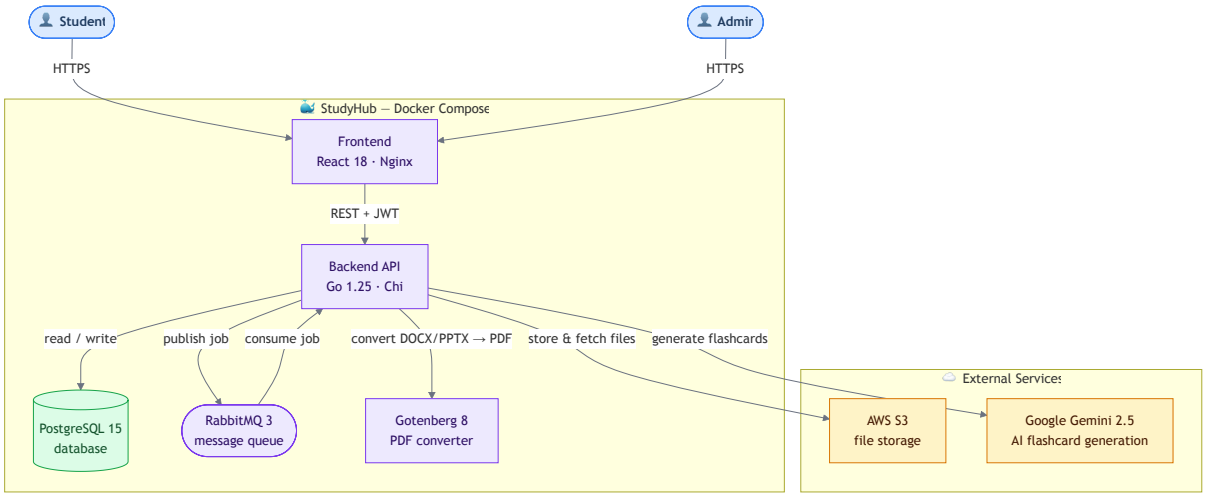
LEVEL 1 System Context

Who uses StudyHub and which external systems does it depend on.



LEVEL 2 Containers

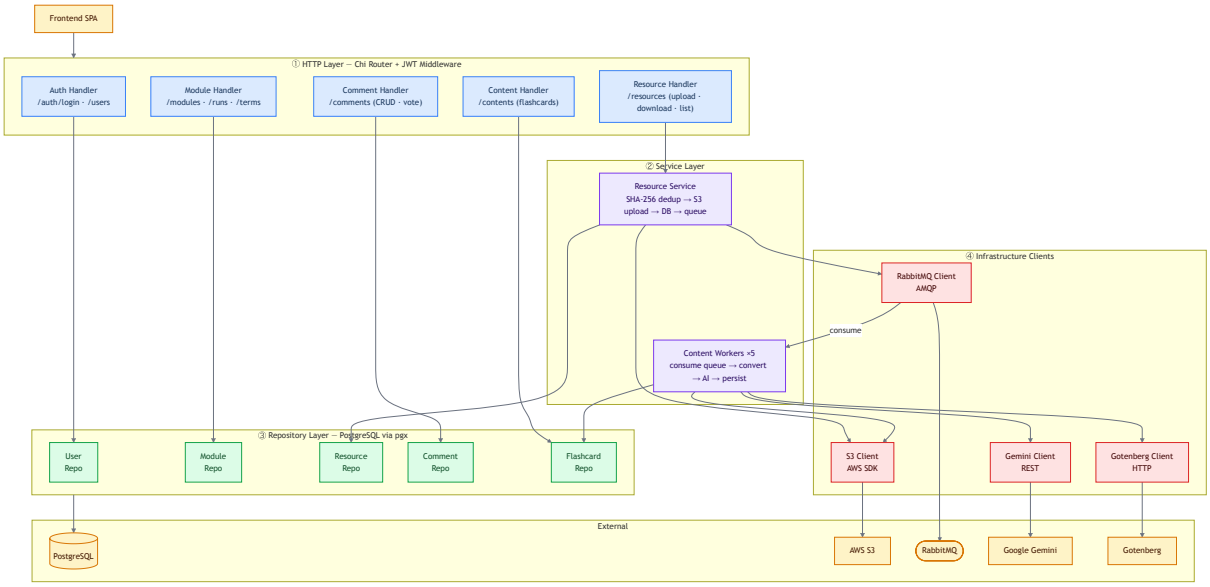
The deployable services that make up StudyHub, split into the synchronous request path and the asynchronous AI processing pipeline.



LEVEL 3

Components — Backend API

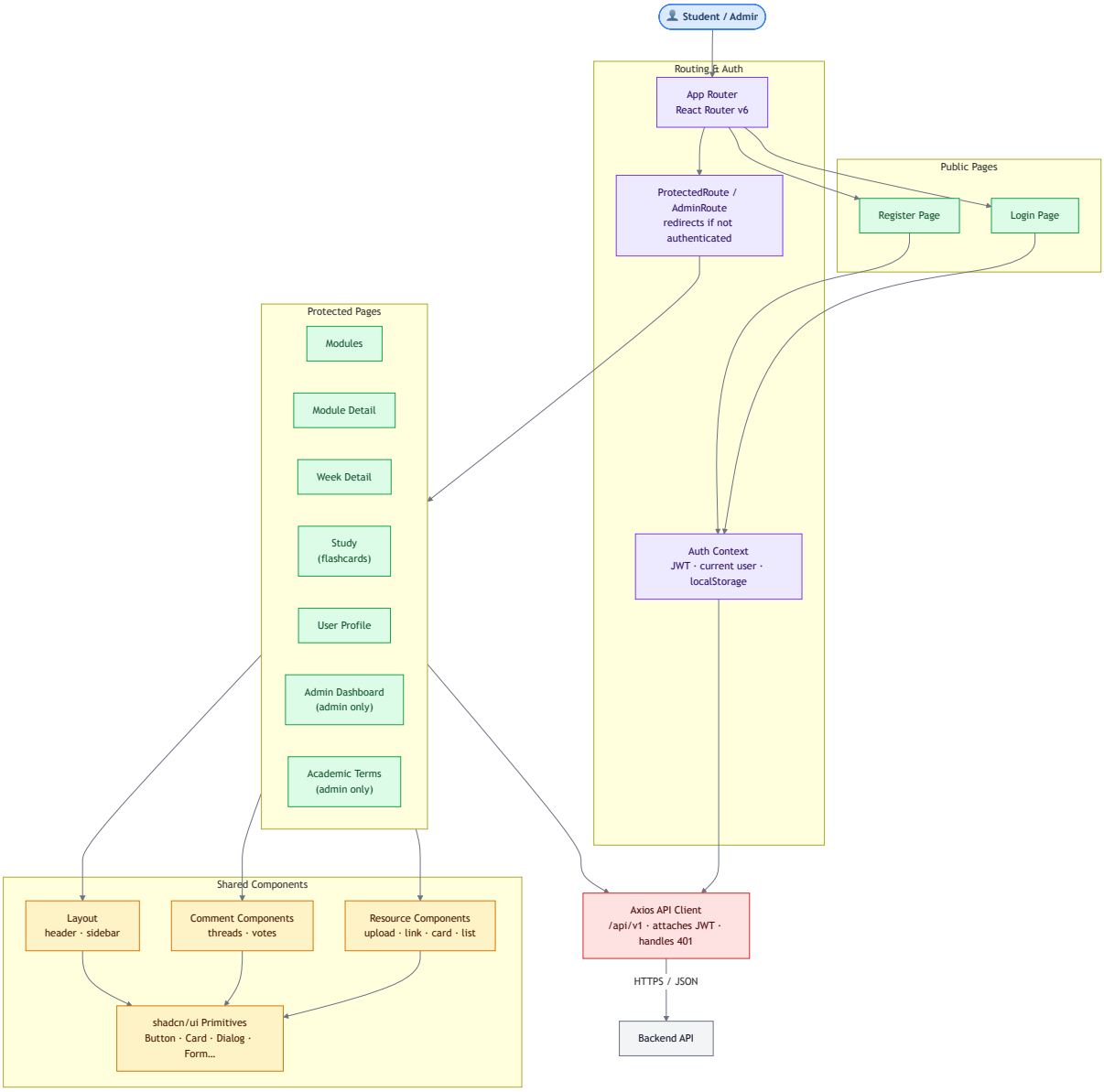
Four clear layers: HTTP handlers receive requests → services apply business logic → repositories talk to the database → infrastructure clients talk to external services.



LEVEL 3

Components — Frontend SPA

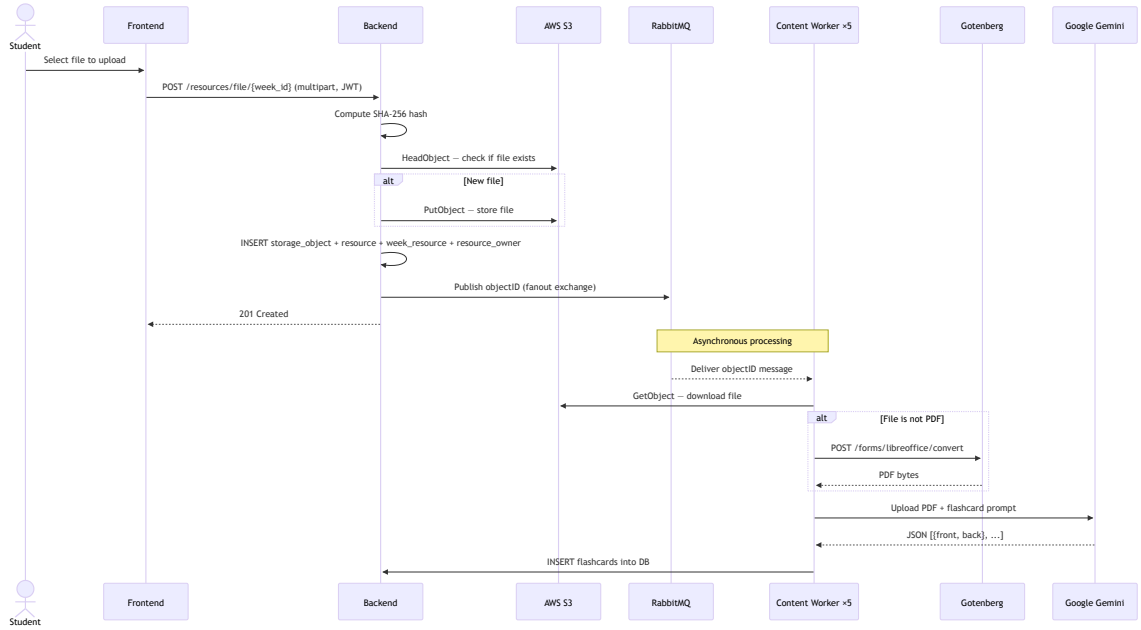
Request flow: Router guards access → pages call the API client → shared components render the UI — all built on shadcn/ui primitives.



FLOW

File Upload + Flashcard Generation

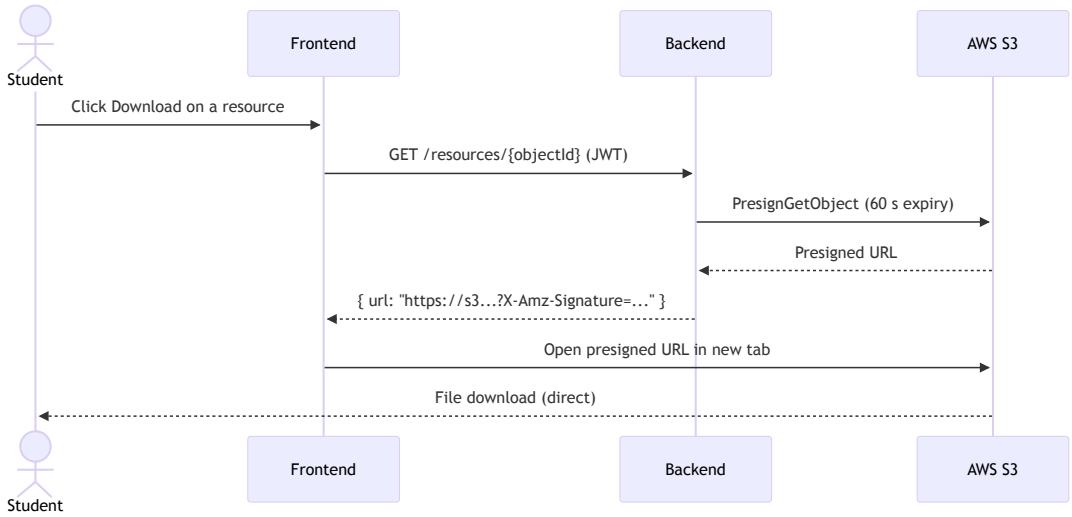
End-to-end flow from a student uploading a file to flashcards being generated by AI.



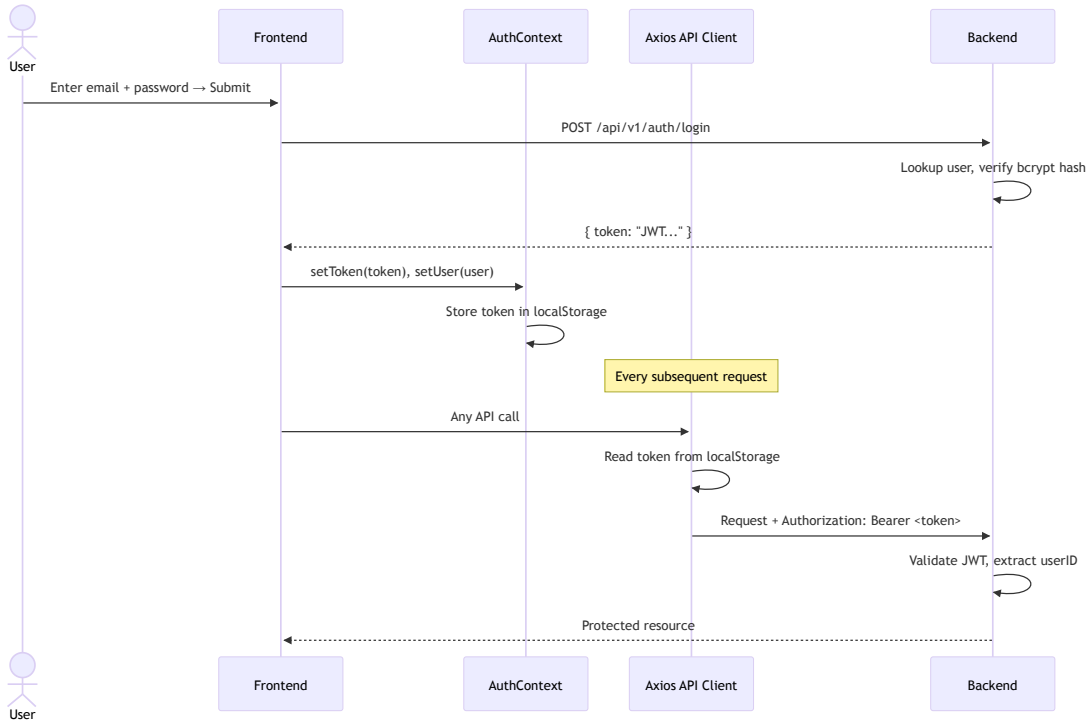
FLOW

Resource Download via Presigned URL

How files are securely served from S3 without exposing credentials.



FLOW

Authentication Flow

How a user logs in and how JWT is propagated to subsequent API calls.